

Manuscript Review

Entropy-Gated Contrastive Decoding

September 15, 2026

Overall assessment and scope

The proposed query-level gate is simple and potentially useful, and the manuscript reports substantial balanced-accuracy improvements for several models. However, the present evidence is not yet reliable enough to support the conclusions. The most urgent problems are an incompatible evaluation denominator for LLaVA-v1.6-Mistral-7B, significance statements inconsistent with the reported counts, and a description of SID that does not match the cited method. Claims of identifying or resolving Visual Inertia also exceed the measurements reported. These require substantive revision before submission, not just copyediting.

This review covers the complete source `latex/sn-article.tex`, all nine supplied figure assets, the appendix threshold table, and the bibliography, including its rendered appearance in the 20-page manuscript PDF. It follows the supplied skill and reference rubric. Locations below use manuscript section, equation, figure, and table numbers; source line numbers refer to that file unless another file is named. No experiment logs, split identifiers, or inference implementation were supplied, so the review distinguishes internal contradictions from hypotheses requiring those artifacts.

Priority technical findings

1. The LLaVA-v1.6 results cannot use the stated common held-out split

Location: Section 4.4, Table 1, Section 5, and Appendix Table A1; source lines 206, 264–269, and 398–402.

Definite internal inconsistency; likely mixed evaluation populations. Assuming the stated unweighted binary metrics and ordinary rounding to two decimals, the first six model blocks in Table 1 are consistent with a common held-out population of 488 positive and 345 negative examples. For example, Qwen2-VL’s baseline gives $241/488 = 49.39\%$ recall and $323/345 = 93.62\%$ specificity. Searching all positive/negative integer denominators summing to 833 yields this pair for each of those six blocks, but *no pair* for the LLaVA-v1.6 block.

Instead, the latter block is reproduced by denominators 975 and 689, summing to the entire stated 1,664-example dataset:

LLaVA-v1.6 regime	Correct positive	Correct negative	Balanced accuracy
Greedy / EGCD / SID	764/975	646/689	86.06%
Universal CD+APC	633/975	595/689	75.64%
VCD	748/975	623/689	83.57%

This compatibility is evidence of a possible full-dataset/held-out mixture, not proof of how the experiments were run. Nevertheless, the stated common-split protocol and the printed numbers cannot both be correct. The identical 86.06 plateau in the tuning table also warrants a provenance check.

Suggested fix: Export the evaluation IDs and integer confusion matrices for every model/regime; rerun the affected results on the frozen 833-item split, then regenerate all dependent figures and conclusions. If results were averaged or filtered, disclose the actual denominator and aggregation instead. Report class counts for both partitions; the held-out population implied by the first six blocks is not class-balanced, so the benchmark’s “balanced by construction” statement in Section 4.1 needs explicit verification rather than being used to equate accuracy with balanced accuracy.

2. The significance analysis conflicts with the reported metrics

Location: Section 5.1, paragraph immediately after Table 1; source line 274.

Definite inconsistency under the stated conventional, unadjusted paired-test interpretation. For Phi-3, the denominators above imply 637 correct answers for EGCD ($349 + 288$) and 643 for universal CD ($371 + 272$). EGCD therefore has slightly higher balanced accuracy but *six fewer* correct answers overall. A pooled McNemar test concerns paired correctness, not the class-reweighted balanced-accuracy difference; its exact version uses discordant-pair binomial counts [1].

The printed $p > 0.99$ cannot be an ordinary unadjusted two-sided McNemar result for these Phi-3 marginals. Enumerating all feasible paired correctness tables with $N = 833$, 637 EGCD successes, and 643 CD successes gives a maximum exact p of about 0.7992. SmolVLM2 similarly implies 540 versus 542 correct answers, with a maximum exact p of about 0.9670, again below 0.99. Conversely, the claim that Qwen2-VL is the *only* significant EGCD improvement over CD ignores LLaVA-1.5: its reported marginals imply 637 versus 477 correct, for which even the largest feasible exact p is below 10^{-11} . These are bounds, not reconstructed actual paired tests.

Suggested fix: Publish discordant-pair counts, the exact test variant, comparison direction, and any multiplicity adjustment. Restrict “only” to a clearly defined subset if that was intended. Report paired uncertainty for the actual balanced-accuracy difference, for example through an appropriately designed stratified paired bootstrap, and account for repeated images. Do not interpret nonsignificance as equivalence or call it “sampling noise” without uncertainty estimates. The reported Qwen2-VL $p = 0.002$ and VCD $p = 0.032$ still need verification from item-level outputs.

3. The described SID baseline is not the cited SID method

Location: Sections 2.2, 4.4, and 5.1; source lines 91, 206, and 274; citation key huo2025self.

Externally verified description error; implementation needs verification. The manuscript describes SID as using a shallow-layer amateur. The cited Self-Introspective Decoding instead uses Context and Text-aware Token Selection (CT²S), retaining low-importance visual tokens after early

decoder layers to construct its disturbed pathway; it is not simply a shallow-layer logit readout [2]. Confusing these procedures changes the baseline being tested and undermines conclusions drawn from its near-zero effects.

Suggested fix: Identify whether the implementation actually performs CT²S. If not, rename the present ablation and evaluate the published SID method separately. Specify the selection layer, retained-token policy, and architecture adaptations. Also give VCD’s noise schedule, perturbation strength, and seed: matching only α and β does not specify either baseline. Correct Section 2.2’s grouping of layer-wise logit contrast with interventions that edit hidden activations; these are distinct operations.

4. First-token uncertainty does not establish a Visual Inertia mechanism

Location: Abstract; Sections 1, 3.1–3.2, 5.2–5.4, and 6.1; particularly lines 65, 69–74, 129, 284, and 309.

Unsupported causal claims. The manuscript measures answer performance and routing, but does not report attention trajectories demonstrating that its evaluated errors arise from attention stagnation, or that EGCD changes that behavior. A statistic measured before the first generated answer token cannot, by itself, establish a process described as grounding decay *during subsequent decoding*. The motivating paper studies token-wise attention dynamics [3]; that evidence does not substitute for a mechanism test in this experiment.

The AUC range 0.62–0.75 appears only in the contribution list, without model-level values, sample counts, uncertainty, or even a definition of which outcome is the positive class. Even if confirmed, error-prediction AUC would not show that entropy specifically detects vision–text conflict, nor that every high-entropy query benefits from CD. Low entropy can accompany a confidently wrong prediction. Similarly, high baseline recall alone does not identify a connector-level causal explanation.

Suggested fix: Reframe the gate as an uncertainty-based heuristic whose mechanistic interpretation is a hypothesis. Supply the AUC analysis and separately measure whether entropy predicts the *benefit of intervention*: CD fixes versus harms, stratified by entropy and label. To retain the stronger claims, measure attention dynamics and relevant image/text counterfactual controls on the evaluated models. Label the attention overlay in Figure 1 as schematic unless it is a documented measurement. Describe z_{cf} as text-only conditional logits, not a proven separable component of multimodal bias.

5. The reported gains do not imply reduced false-positive hallucination

Location: Section 4.3, final sentence; Section 5.2; Section 6.1, source lines 202, 294, and 338.

Definite numerical conflict plus unsupported interpretation. Section 6.1 says the biased models improve “without the specificity loss that universal CD leaves behind.” Relative to greedy decoding, however, EGCD lowers Qwen2-VL specificity from 93.62% to 86.38% and Phi-3 specificity from 91.59% to 83.48%: losses of 7.24 and 8.11 percentage points. On the manuscript’s own definition of hallucinated Yes answers, false-positive rates rise from 6.38% to 13.62% and from 8.41% to 16.52%, respectively. The method improves balanced performance by recovering positives, not by uniformly reducing hallucination errors.

Suggested fix: State that EGCD reduces the specificity *cost relative to universal CD*, while still sacrificing specificity relative to greedy decoding. Carry that qualification into the abstract and conclusion. Replace “Only a method that restores visual grounding moves both terms at once”

with a description of what recall and specificity measure: neither simultaneous improvement nor a balanced-accuracy gain is a causal test of visual grounding. In Section 5.1, replace the exact claim that EGCD “matches universal CD+APC wherever that method helps”: Qwen2.5-VL reports 82.78% versus 82.85%, not equality.

6. Gate-direction prose and a central figure caption contradict the rule

Location: Sections 5.3–5.4, Figure 6 caption, Appendix Table A1; source lines 311, 316, and 321.

Definite mathematical and cross-reference inconsistencies. Section 3.2 defines intervention by $H(Y_1) \geq \tau$. Consequently, for fixed examples, raising τ can only *decrease* or preserve the number routed. Section 5.4 instead attributes the rise toward Qwen2-VL’s peak as τ increases to sending more uncertain queries to System 2. The improvement must come from excluding lower-entropy interventions, not increasing the intervention count.

Figure 6’s caption says EGCD tracks the baseline “everywhere.” Its own LLaVA-1.5 curve and Table A1 rise from 61.47% at $\tau = 0.55$ to 75.56% at $\tau = 0.70$; the corresponding baseline plateau is 75.56%. That is a 14.09-point low-threshold loss, not baseline preservation throughout. A nearly flat Idefics2 performance curve likewise does not establish an idle gate: many interventions could leave the argmax unchanged.

Suggested fix: Explain the sweep in terms of *fewer, more selectively retained* interventions as τ increases. Limit baseline-preservation statements to the selected thresholds on the tested split. Plot routing fraction versus τ separately from accuracy and report per-query output agreement when claiming exact preservation. Replace “tracks the greedy baseline everywhere” with “recovers the greedy baseline at sufficiently high thresholds for the two LLaVA models.”

7. One performance plot disagrees with the tabulated equality

Location: Figure 2, rightmost model; Table 1 and Figure A1 (`latex/fig2.png` and `latex/fig8.png`).

Definite figure–table conflict. Figure 2 places the LLaVA-v1.6 SID marker visibly above EGCD. Table 1 and Figure A1 give both 86.06%, with greedy decoding equal as well. A visual difference here contradicts the principal idle-gate comparison, independently of the denominator problem.

Suggested fix: Regenerate all figures from a single verified results table after correcting Finding 1. Assert equality of plotted coordinates when the underlying values are identical. Maintain explicit tuning/held-out labels: Figure A2 correctly shows tuning gains such as 6.43 points for Qwen2-VL, not its held-out 9.39-point gain. Amend Section 4.4’s statement that *every* number is held-out, since Figures 5–7 and the appendix deliberately report tuning results.

8. Entropy units, vocabulary support, and decoding temperature need specification

Location: Equation (2), Section 3.2, Figure 1, and Section 4.4; source lines 122–129 and 206.

Implementation ambiguity. Equation (2) sums over the full vocabulary, while the explanation and diagram present a Yes/No distribution. These are not interchangeable. With natural logarithms a renormalized binary distribution has maximum entropy $\log 2 \simeq 0.6931$, so thresholds of 0.70 or 0.71

cannot activate such a gate. Full-vocabulary entropy can exceed that bound and is mathematically compatible with those thresholds; the thresholds are not intrinsically erroneous.

The log base is unspecified. “Temperature zero” must describe greedy token selection, not the distribution used to compute entropy: the zero-temperature limit is degenerate apart from ties. The assertion that the first emitted token is always the answer also needs the exact prompt and tokenizer-level evidence.

Suggested fix: State whether the implemented statistic is unnormalized full-vocabulary entropy in nats, evaluated from unscaled logits before greedy selection. If another definition was used, rewrite Equation (2) and retune consistently. Document chat templates, leading whitespace/special-token handling, Yes/No tokenization, and response parsing. For half-precision model inference, compute log-softmax and the entropy reduction in adequate precision and define zero-probability handling to avoid numerical NaNs.

9. Bind the APC candidate consistently and complete the generation algorithm

Location: Equations (1), (4), and (5); Algorithm 1; source lines 114, 139–147, and 153–174.

Definite notation defect and incomplete pseudocode. Equation (4) binds the set member as y but tests $P_{\text{std}}(y_t \mid \dots)$; the maximum similarly binds w while evaluating w_t . Literally, the membership test is not a function of the bound candidate. Equation (1) also describes $z_{\text{std}}(y_t)$ as a vocabulary vector while later equations use it as a scalar token logit.

Suggested fix: Define the vector and its components separately and write, with the context suppressed,

$$P_{\text{std}}(y) = \frac{\exp z_{\text{std}}(y)}{\sum_{w \in V} \exp z_{\text{std}}(w)}, \quad V_{\text{head}} = \{y \in V : P_{\text{std}}(y) \geq \beta \max_{w \in V} P_{\text{std}}(w)\}.$$

Specify $0 < \beta \leq 1$ and $\alpha \geq 0$, with the existing numerical choices unchanged. The contrastive sign and normalization are otherwise coherent: within the head, the normalized distribution is proportional to $P_{\text{std}}(y)^{1+\alpha}/P_{\text{cf}}(y)^\alpha$. The rule explicitly handles $H = \tau$ via System 2; there is no missing equality branch.

Algorithm 1 does not initialize or append to either returned sequence, and only System 2 checks EOS. Initialize a shared generated prefix, append each selected token, stop on EOS in both branches, and return that prefix. Feed the same selected prefix to both System 2 streams, but maintain their distinct caches. Reuse the initial expert logits for the first token instead of recomputing them. If evaluation generates exactly one answer token, state that restriction and simplify the algorithm accordingly.

10. Calibration, split independence, and cost claims need narrower reporting

Location: Sections 1, 3.2, 4.4, 5.4, and 6; source lines 69, 129, 206, and 333–342.

Protocol omissions and unsupported generalization. The per-model threshold is selected using labeled tuning accuracy. “Training-free” can describe frozen weights, and “zero-shot” can describe prompting, but the complete procedure is not calibration-free or independent of labeled task data. Appendix Table A1 also contains tied maxima for the LLaVA models without an explicit tie-breaking rule. Per-model calibration, rather than architecture recognition alone, contributes to the reported safe operating points.

The split description omits the random seed, IDs, and whether partitioning is by image or query. The official relation query file contains multiple questions for the same image [4], so query-level splitting does not establish image-independent evaluation. Shuffled-routing comparisons report neither repetitions nor uncertainty, leaving the stability of those control values unknown.

Suggested fix: Describe “zero-shot prompting with supervised threshold calibration on a disjoint tuning split.” Release split IDs, class/image counts, grouping policy, selection objective, and deterministic tie-breaking rule. Report repeated rate-matched shuffled controls and quantify uncertainty; assess cross-image or cross-dataset transfer before generalizing the architectural boundary.

Finally, the entropy reduction is inexpensive but not literally free, and Algorithm 1 runs a counterfactual forward evaluation at *each generated token* of a routed query, not just once for an arbitrary-length response. State the actual output-length regime, prefill/cache reuse, and measured latency or forward-pass cost. Routing percentages alone do not validate claims about total inference cost or fluency.

Reference integrity and meaningful editorial findings

Verified bibliographic metadata errors

Location: `latex/references.bib`; keys listed below, and manuscript References [16], [19], [27], and [41].

Definite externally verified errors. These are attribution errors, not stylistic preferences:

- `wang2025contextfocus` assigns ContextFocus to Wang and Chen in 2025. Its arXiv record names Nikhil Anand, Shwetha Somasundaram, Anirudh Phukan, Apoorv Saxena, and Koyel Mukherjee, with first submission on January 7, 2026 [5].
- `huo2025self` contains incorrect given names throughout its author list. The proceedings list Fushuo Huo, Wenchao Xu, Zhong Zhang, Haozhao Wang, Zhicheng Chen, and Peilin Zhao [2].
- `liu2023visual` lists “Wu, Yong”; the actual author is Qingyang Wu [6].
- `gong2026attention` should list Boyang Gong, Yu Zheng, Fanye Kong, Jie Zhou, and Jiwen Lu, rather than the differing names currently present [3].

Suggested fix: Import verified metadata from the corresponding primary records, preserving existing citation keys unless a deliberate global change is made. Check the remaining bibliography systematically; these spot checks do not certify its other entries. Recheck citation support as well as names: ContextFocus concerns contextual faithfulness in language models, so its use as evidence for visual-model behavior needs qualification [5].

The bibliography style corrupts arXiv identifiers

Location: Rendered References, pp. 17–20; `latex/references.bib` journal fields; `latex/sn-basics.bst`, lines 190 and 1190–1193.

Definite formatting defect affecting reference retrieval. ArXiv identifiers are embedded in the `journal` field, which this style passes through `remove.dots`. Consequently, the PDF prints identifiers

such as `arXiv:240414219` instead of `arXiv:2404.14219`. This was checked in the rendered page, not inferred solely from text extraction.

Suggested fix: Move the identifier into a style-supported field that preserves it, such as a suitable `note` field, with `journal = {arXiv preprint}`, or use verified DOI/eprint formatting supported by the template. Rebuild and inspect the reference list to confirm the decimal separator survives. Do not remove journal-abbreviation handling globally just to repair identifiers.

Bounded proofing sweep

The supplied `scripts/proofing_scan.py` was run on the manuscript PDF, source, and BibTeX file with an 80-hit cap. Hits were spot-checked. The PDF’s ellipsis and semicolons before CHAIR, MME, HaluSearch, VCD, and Phi-3 are not errors. No genuine duplicated-punctuation defect was found. The following concrete corrections remain:

- **Appendix A opening paragraph, line 373:** the scan flags “; Recall and Specificity”. Use “; recall and specificity” as ordinary metric names, or begin a new sentence if retaining sentence-initial capitalization.
- **Section 1, contribution 2, line 74:** “reads a statistic, the standard pass computes anyway” contains a comma separating the noun from its defining clause. Use “reads a statistic that the standard pass already computes”; qualify the cost as in Finding 10.
- **Section 3.2, line 129:** plural “Queries” becomes singular “answers it”. Use “System 1 answers them with standard greedy decoding” or make the whole sentence singular.
- **Figure 1 footer:** the printed tuned-threshold interval $\tau \in [0.650, 0.710]$ excludes Phi-3’s selected 0.62. Change it to $[0.620, 0.710]$ if describing selected thresholds, or $[0.550, 0.750]$ if describing the search grid.
- **References, pp. 17–20:** unprotected titles render proper names as “Qwen2.5-vl”, “Spatialvlm”, “Dola”, and “Smolvlm”. Protect the corresponding tokens in BibTeX, for example `{Qwen2.5-VL}`, `{SpatialVLM}`, `{DoLa}`, and `{SmolVLM}`. This preserves names rather than imposing title-case styling.

The title page also still contains placeholder authors, example email addresses, and an unfilled affiliation. Replace them before submission; they are not valid publication metadata.

Coverage record and remaining verification

The technical pass checked all eight numbered equations, Algorithm 1, the appendix grid and selected maxima, sign conventions, candidate binding, probability normalization, entropy units, equality branches, and implementation details. No inverse-trigonometric or coordinate-transform expressions occur, so quadrant and coordinate-frame checks are inapplicable rather than omitted. The printed balanced accuracies agree with the means of recall and specificity to displayed rounding precision; the major numerical defects concern populations, inference, and cross-artifact consistency rather than that averaging formula.

The required notation/terminology check tracked X_v and X_t from Section 3.1 (visual input and text prompt) into the text-only pathway; $Y_{<t}$ from the generated prefix into Algorithm 1; $H(Y_1)$ from full-vocabulary first-token entropy into the binary explanation and conflict claims; τ and τ^* from a routing boundary into the tuning-selected operating point; and V_{head} from plausible-token membership into the algorithm. The concrete conflicts are the vocabulary/support ambiguity, reversed threshold-direction explanation, inconsistent candidate indexing, and the shift from empirical uncertainty to asserted causal conflict. No source/observer or initial/final angle terminology occurs.

Needs verification beyond the supplied artifacts: actual evaluation IDs and paired outputs; correctness of the baseline implementations; the AUC and randomized-routing analyses; image overlap between partitions; and claims that the evaluated data or annotated objects are absent from the training corpora of all seven models. AMBER’s original collection rationale concerns image sourcing and the models considered in that work [7]; it does not establish a contamination guarantee for every later checkpoint. The remaining literature metadata and the precise borrowed Visual Inertia ablation values also need primary-source checking. These unresolved points are not presented as proven experimental misconduct or fabricated results.

Highest-priority revision order

1. Reconcile evaluation populations and release the integer counts and split IDs before interpreting any seven-model comparison.
2. Recompute paired inference for the stated metric, verify SID/VCD, and regenerate tables and figures from one results source.
3. Specify entropy, prompting, prefix/caching behavior, threshold selection, and actual inference costs sufficiently for reproduction.
4. Restrict mechanistic and hallucination-reduction claims to what the corrected evidence supports; then repair references and the bounded copy defects.

References

- [1] Statsmodels developers. *statsmodels.stats.contingency_tables.mcnemar*. Official API documentation, sections on the exact binomial test and returned p -value. Consulted September 15, 2026.
- [2] Fushuo Huo, Wenchao Xu, Zhong Zhang, Haozhao Wang, Zhicheng Chen, and Peilin Zhao. *Self-Introspective Decoding: Alleviating Hallucinations for Large Vision-Language Models*. ICLR 2025; arXiv:2408.02032v3, Sections 4.1–4.2. Primary proceedings and arXiv text consulted.
- [3] Boyang Gong, Yu Zheng, Fanye Kong, Jie Zhou, and Jiwen Lu. *Attention at Rest Stays at Rest: Breaking Visual Inertia for Cognitive Hallucination Mitigation*. arXiv:2604.01989v3, 2026. Author record and abstract consulted.
- [4] AMBER authors. Official repository, `juncywang0410/AMBER`, `data/query/query_discriminative-relation.json`. For example, query IDs 13557 and 13558 use the same image. Consulted September 15, 2026.
- [5] Nikhil Anand, Shwetha Somasundaram, Anirudh Phukan, Apoorv Saxena, and Koyel Mukherjee. *ContextFocus: Activation Steering for Contextual Faithfulness in Large Language Models*. arXiv:2601.04131v2; first submitted January 7, 2026.

- [6] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. *Visual Instruction Tuning*. NeurIPS 2023; arXiv:2304.08485v2. Author record consulted.
- [7] Junyang Wang et al. *AMBER: An LLM-free Multi-dimensional Benchmark for MLLMs Hallucination Evaluation*. arXiv:2311.07397v2, Section 3.1, image collection rationale.